

The glossary provides definitions for key words and terms used in the training package.

Abatement: a reduction in atmospheric GHGs through emissions avoidance or removal and storage of carbon from the atmosphere.

Abatement estimates: estimates of abatement, which may include estimates of potential abatement, or estimates of abatement that has been achieved.

Additionality: the extent to which emissions avoidance or carbon storage would be unlikely to occur in the ordinary course of events (i.e. in the absence of an incentive such as that provided by a carbon crediting scheme). Additionality ensures carbon credits used to offset emissions represent genuine reductions in GHG emissions.

Australian Carbon Credit Unit (ACCU): a unit issued under the Carbon Credits (Carbon Farming Initiative) Act 2011, equal to 1 tonne of carbon dioxide equivalent emissions (as Topic 5 explains).

Australian Carbon Credit Unit (ACCU) Scheme: an Australian Government scheme that offers landholders, communities and businesses the opportunity to undertake projects in Australia that avoid releasing GHG emissions or remove and store carbon from the atmosphere. It is enacted through the Carbon Credits (Carbon Farming Initiative) Act 2011 and the Carbon Credits (Carbon Farming Initiative) Rule 2015 (as Topic 5 explains).

Afforestation: establishing a forest on land where there was no previous forest (as Topic 2.3 explains).

Agroforestry: planting and maintaining trees and shrubs around or on agricultural land, integrating trees and shrubs with crops and livestock (as Topic 2.3 explains).

Australian National Registry of Emissions Units: a secure system administered by the Clean Energy Regulator that tracks the location and ownership of ACCUs (as Topic 5.2 explains).

Anaerobic (pond): A small, deep primary effluent treatment pond where bacteria break down organic matter in the absence of oxygen, releasing methane and carbon dioxide (as Topic 2.3 explains).

Baseline: An estimate of GHG emissions and carbon storage for a nominated year or years, which provides a starting point for comparing changes in emissions and carbon storage over time.

Biodiversity: the variety of genetic information in plants, animals and microorganisms, of species and of habitats, ecological communities and ecological processes.

Biogas: a mix of methane and other gases generated by anaerobic digestion (microorganisms breaking down materials) of food, crop waste and manure.

Biomass: the total mass of living organisms present in an area, ecosystem, environment or a category of organisms.

Blue carbon: the carbon stored by plants and soils of coastal ecosystems that support mangroves, tidal marshes and seagrasses, prevalent along most coastlines in the world, that can capture and store large amounts of carbon in their soils, roots and plants (as Topic 2.3 explains).

Carbon: a non-metallic element found in all animals and plants and in substances such as coal and oil. When carbon is burned it produces carbon dioxide and carbon monoxide.

Carbon cycle: the flow of carbon in various forms (such as carbon dioxide) through the atmosphere, land and oceans (as Topic 1.7 explains).

Carbon dioxide equivalent (CO₂-e): a unit based on the global warming potential of different GHGs compared to carbon dioxide. For example, in 2024, one tonne of methane released into the atmosphere is estimated to contribute as much to climate change as 28 tonnes of carbon dioxide. Therefore, one tonne of methane equals 28 tonnes of CO₂-e.

Carbon farming: changing agricultural practices or land management to increase the amount of carbon stored in the soil and vegetation and to reduce GHG emissions from agricultural production or land management.

Carbon footprint: the amount of GHGs emitted minus the amount of carbon stored, for example by a farm, region or country.

Carbon neutral: a state of emissions and absorption of GHGs from the atmosphere being in balance (as Topic 1.8 explains).

Carbon service provider: a business that offers services to help establish and manage carbon farming activities, including ACCU Scheme projects, in exchange for a fee (for example, a payment or some of the ACCUs issued for the project). They are also known as a carbon project aggregator or developer (as Topic 4.3 explains).

Carbon sequestration: the process of removing carbon from the atmosphere and storing it in trees, soils, geologic formations, oceans and in engineered technologies.

Carbon storage: another term for carbon sequestration.

Caring for Country: activities by First Nations people that draw on laws, knowledge and customs inherited from ancestors and ancestral beings to ensure the continued health of lands and seas with which they have a traditional attachment or relationship.

Clean Energy Regulator: a statutory authority responsible for administering legislation that will contribute to a reduction in Australia's GHG emissions. It has administrative responsibilities for the National Greenhouse and Energy Reporting Scheme, ACCU Scheme, Australian National Registry of Emissions Units, Safeguard Mechanism and Renewable Energy Target.

Climate Active: an Australian Government program, administered by DCCEEW, that encourages business decarbonisation through certifying credible voluntary climate action.

Co-benefit: the benefits in addition to reducing GHG emissions and storing carbon. These are benefits for farmers and land managers, the environment and communities (as Topics 1 and 2 explain).

Country: the First Nations concept of everything within a cultural landscape, including the land and sea; the plants and animals within them; the history, culture and traditions associated with them; and the connections between people and the landscape. Country is a distinct geographic, cultural and ecological space common to a specific Indigenous people, group of peoples or local community. Tenure is held collectively – whether formally recognised legally or otherwise – and resource definition and use and cultural practice are governed within a common property context.

Crediting period (ACCU Scheme): the period during which an ACCU Scheme project can generate ACCUs: generally 7 years for emissions avoidance projects and 25 years for carbon storage projects (as Topic 5.3 explains).

Cultural burning: deliberate burning of the landscape authorised and led by the Traditional Owners of that Country for purposes including ceremonies, protecting cultural and natural assets, reducing fuel, healing Country's spirit and managing and regenerating food, fibre, medicines and plant and animal habitat (as Topic 1.4 explains).

Ecosystem: a community of living things and the non-living environment functioning together as an ecological system.

Eligible interest holder (ACCU Scheme): a person or organisation with a legal interest in the land on which an ACCU Scheme project will be run, including landholders, banks or mortgagees, state and territory Crown Lands Ministers and registered native title bodies corporate (as Topic 5.3 explains).

Emissions: outputs and discharges, as in the introduction of chemicals or particles into the atmosphere, usually used in relation to GHG emissions.

Enteric fermentation: the process by which microbes in the rumen of ruminant livestock produce methane as their feed ferments, which the animal then belches (as Topic 1.6 explains).

Fallow land: land rested from cropping for a time so the soil can maintain its capacity to support plant growth and high crop yields.

First Nations: a generic term for Indigenous peoples. It is not specific to Australian Indigenous peoples and can be applied to describe Indigenous peoples from other countries.

Forward abatement estimate (ACCU Scheme): an estimate of the expected total amount of carbon stored, or GHG emissions avoided, by an ACCU Scheme project in tonnes of carbon dioxide equivalent. A project proponent provides the estimate when applying to register an ACCU Scheme project.

Free, Prior and Informed Consent (FPIC): FPIC is a decision-making process and a framework for ensuring First Nations people are properly engaged in any decision that may affect their lands, territories or livelihoods and that outside organisations engage with First Nations people in a culturally appropriate way.

Global warming potential: a value used to compare the ability of different GHGs to trap heat in the atmosphere, with carbon dioxide used as a benchmark. For example, in 2024, methane's global warming potential (GWP) value was 28. This means that one tonne of methane released into the atmosphere is estimated to contribute as much to climate change as 28 tonnes of carbon dioxide (as Topic 1.8 explains).

Greenhouse gas (GHG): a gas that traps heat in the atmosphere. The main GHGs are carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs) and sulfur hexafluoride (SF₆).

Insetting: undertaking activities that reduce or avoid emissions, or store carbon within a value chain, which may comprise a farm and its supply chain, and counting the emissions reductions or carbon storage towards the operation's total emissions.

Life cycle assessment: an assessment of the environmental aspects and potential impacts of a product, process or service by compiling an inventory of relevant energy and material inputs and environmental releases, evaluating the potential environmental impacts of those inputs and releases and interpreting the results to inform decision-makers (as Topic 3.5 explains).

Methane: one of the main GHGs produced by agricultural activities: methane emissions, mainly by beef and dairy cattle and sheep, but also pigs, account for most of Australia's agricultural emissions (as Topic 1.6 explains).

Method (ACCU Scheme): a legislated set of requirements and rules for running an ACCU Scheme project. Each method specifies the activities you can conduct, how to measure carbon abatement and monitoring, record-keeping and reporting requirements (as Topic 5 explains).

National Plantation Inventory (NPI) region: regions in which tree plantations are grown primarily for timber production. The regions are defined in National Plantation Inventory reports prepared by the Australian Bureau of Agricultural and Resource Economics and Sciences.

Nature positive: a term used to describe circumstances where nature – species and ecosystems – is being repaired and is regenerating rather than declining.

Net zero emissions: taking steps to reduce GHG emissions as much as possible and to use carbon storage to balance remaining emissions, over a specified period (as Topic 1.8 explains).

Nitrogen cycle: a biogeochemical process that begins with the conversion of nitrogen in the atmosphere into ammonia by bacteria in the soil (nitrogen fixation) and its transformation into nitrites and nitrates (nitrification) that plants use to grow and develop (assimilation). When plants and the animals that eat them die, organic matter breaks down, releasing nitrogen back into the soil as ammonium (ammonification). Finally, bacteria convert the nitrates back into atmospheric nitrogen gas (denitrification) (as Topic 1.6 explains).

Nitrous oxide: one of the main GHGs produced by agricultural activities. Sources include burning crop residues and using fertilisers (as Topic 1.6 explains).

Offsetting: the buying and cancelling (also referred to as retiring) of ACCUs or other eligible carbon credits by an organisation to compensate for the emissions it produces.

Permanence period (ACCU Scheme): the period over which carbon stored by an ACCU Scheme project must be maintained. A period of either 25 or 100 years can be adopted (as Topic 5.3 explains).

Precision agriculture: the use of technology and data analysis to optimise various aspects of agricultural production (as Topic 2.3 explains in relation to soil organic carbon). It is also called 'precision farming'.

Project proponent (ACCU Scheme): the party with the legal right to carry out an ACCU Scheme project. A project proponent is responsible for carrying out the project, is issued all ACCUs and is legally responsible for meeting all obligations under the law (as Topic 5.3 explains).

Relinquish (ACCU Scheme): to return ACCUs to the Clean Energy Regulator. Project proponents may voluntarily relinquish ACCUs in some circumstances. There may also be circumstances where the Clean Energy Regulator requires ACCUs to be relinquished.

Revoke (ACCU Scheme): to withdraw a registered ACCU Scheme project. A project proponent can elect to revoke their project. They may need to meet certain obligations associated with the revocation.

Reforestation: establishing a forest on land where there has previously been a forest (as Topic 2.3 explains).

Risk of reversal buffer (ACCU Scheme): a discount applied to the number of ACCUs issued to carbon storage projects to protect the ACCU Scheme against the potential loss of carbon and other risks that can't be managed by other permanence arrangements. The risk of reversal buffer reduces the ACCUs issued for each reporting period by 5% for vegetation and soil carbon projects (as Topic 5.2 explains).

Savanna: land characterised by a tropical or sub-tropical vegetation formation with continuous grass cover occasionally interrupted by trees and shrubs.

Scope 1 emissions: emissions released into the atmosphere from operations a business owns or controls, such as methane from livestock digestion and manure management and nitrous oxide from fertiliser use on a farm. Scope 1 emissions are also called direct emissions (as Topic 1.6 explains).

Scope 2 emissions: emissions released into the atmosphere from generating electricity, steam, heat or cooling a business buys (for example, emissions from burning coal to produce electricity a farm uses). Scope 2 emissions are also called indirect emissions (as Topic 1.6 explains).

Scope 3 emissions: all indirect (off-farm) emissions (other than scope 2 emissions) that occur in the business' value chain. For a farm business, this could include upstream emissions from producing and transporting fertilisers and pesticides and downstream emissions from transporting, processing and consumption of the farming business' products, including waste disposal (as Topic 1.6 explains).

Sink (carbon): places where carbon dioxide that is captured and removed from the atmosphere is stored, including trees, other vegetation, soil, geological formations (via carbon capture and storage), long-lived products (via carbon capture and use) and minerals.

Soil organic carbon (SOC): all living and dead organic material — plants, soil organisms and animal materials — in the soil in various stages of decomposition, but not the fresh, undecomposed organic material on the surface (as Topic 1.7 explains).

Spot price (ACCU Scheme): the price of ACCUs bought for immediate delivery (as Topic 5.5 explains).

Supply chain: the sequence of processes involved in producing and distributing a commodity. For an agricultural commodity, this includes the inputs needed to produce it, the production activities, and the subsequent processes for distribution and sale.

Sustainability loan: a loan from a government or private financier for equipment or operations to improve sustainability, including of farming systems and natural resources, for investment in areas such as drought and climate change preparedness, biosecurity and net zero operations.

Tillage: the manipulation of the soil into a desired condition by mechanical means (as Topic 2.3 explains).

Traditional Owner: a First Nations owner of their traditional Country, as determined through the purchase of freehold, as granted by a government or as determined through the native title process.

Value chain: the range of value-adding activities involved in producing and selling a commodity.

Water cycle: the continuous movement of water above and below the ground and in the atmosphere.

Whole of Country plan: overarching, long-term visions developed by Traditional Owner groups that set out clear goals and priorities, principles of engagement and measures of success in Caring for Country.

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